

AIG130 – Lab 2

Objective

This lab focuses on understanding the practical applications, challenges, and benefits of using cloud services for machine learning in real-world scenarios.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Assignment

Select a real-world case where cloud services are employed for machine learning. You will conduct a thorough research and analysis of the case, and then present your findings, focusing on specific aspects like technology, implementation, outcomes, and the broader implications.

Suggested Steps

1. Case Selection:
 - a. You can choose a case from a range of sectors such as healthcare, finance, retail, or technology.
 - b. The case should involve the use of cloud services for machine learning.
2. Research:
 - a. **Technology Analysis:** Investigate the specific cloud services and machine learning models used in the case. Understand why these choices were made over others.
 - b. **Implementation Strategy:** Examine how the cloud services were integrated and the ML models were deployed. Focus on scalability, data handling, and computational requirements.
 - c. **Challenges and Solutions:** Identify any challenges faced during implementation and how they were overcome.
3. Analysis:
 - a. **Effectiveness:** Assess the effectiveness of the ML solution in achieving its objectives.
 - b. **Cost-Benefit Analysis:** Consider the economic aspects, including cost savings or revenue generation.
 - c. **Impact Assessment:** Evaluate the broader impact of the solution, such as improvements in efficiency, customer satisfaction, or innovation.

Deliverables

Report: Submit a PDF report covering all three tasks. Include:

- An introduction and conclusion.
- Sections for each task with headings.
- Visual elements (charts, tables) where applicable.

Presentation: Prepare and deliver a presentation of your findings in class. The presentation is also required to be submitted in Blackboard. It can be as PDF, PPTX, Link, etc.

- Duration: Up to 15 minutes per group.
- Cover key points and findings from the report.
- Encourage questions and discussions.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?
3. What steps would you take to decide on whether to use a cloud service for your ML project? Which criteria would you consider when choosing a cloud platform (e.g. AWS, GCP, Azure)?

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 4%
2. Individual reflection: 2%
3. Presentation file/link: 2%

Assessment Criteria

- Quality and depth of research and analysis.
- Clarity and organization of the report.
- Effectiveness of the presentation.
- Thoughtfulness of the individual reflection.